

Why Measure Theory

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1 Introduction: Motivation

1.1 Definitions

I will begin with the definition of Riemann Integration.

Definition 1 (Reimann Integration). Let f be a bounded function $f: [a, b] \rightarrow \mathbb{R}$. Let P be a partition of the domain of the form

$$P = \{x_0, x_1, \dots, x_n\}$$

Such that

$$a = x_0 < x_1 < \dots < x_n = b$$

If for every $\epsilon > 0$ there exists a partition P_ϵ such that $U(f, P_\epsilon) - L(f, P_\epsilon) < \epsilon$ then $f(x)$ is Riemann Integrable with $\int f = U(f) = L(f)$.

Where $U(f, P)$ is the upper sums

$$U(f, P_\epsilon) = \sum_{k=1}^n (x_k - x_{k-1}) \sup_{[x_{k-1}, x_k]} f$$

and $U(f)$ is inf over all paritions P

$$U(f) = \inf_P U(f, P)$$

Where $L(f, P)$ is the lower sums

$$L(f, P_\epsilon) = \sum_{k=1}^n (x_k - x_{k-1}) \inf_{[x_{k-1}, x_k]} f$$

and $L(f)$ is sup over all paritions P

$$L(f) = \sup_P L(f, P)$$

(Shorten this section its in introduction to show that riemann integration fails here)

1.2 Riemann Failure

1.2.1 Function

Let the sequence of rational numbers $\{r_1, r_2, \dots\}$ enumerate the rationals on $[0, 1]$.

Let $f_n(x)$ be a series of functions of the form

$$f_n(x) = \begin{cases} 1 : & x \in \{r_1, r_2, \dots, r_n\} \\ 0 : & x \notin \{r_1, r_2, \dots, r_n\} \end{cases} \quad (1)$$

1.2.2 Upper and Lower Sums

Fix $n \in \mathbb{N}$.

Take the partition P_n such that $(x_k - x_{k-1}) \leq \frac{\epsilon}{4n}$ for all consecutive pairs of

points in P_n .

The total number of intervals in which there are rational numbers is $2n$; since there are n rationals where each can be in at most 2 intervals at once (Footnote this maybe) (if $r_k = x_k$ for instance then $[x_{k-1}, x_k]$ and $[x_k, x_{k+1}]$ both contains r_k).

This gives

$$U(f_n, P_n) = \left(\frac{\epsilon}{4n}\right)(2n)(1) + (1)(0) = \frac{\epsilon}{2}$$

Where each length $(x_k - x_{k-1}) = \left(\frac{\epsilon}{4n}\right)$, there exists at most $2n$ of these intervals, each with $\sup_{[x_k - x_{k-1}]} f = 1$. All other possible intervals have a max combined length of 1, each with $\sup_{[x_k - x_{k-1}]} f = 0$.

Since $\mathbb{I}$ is dense in $\mathbb{R}$, between any two real numbers there exists an irrational. Meaning on any interval $[x_{k-1}, x_k]$, $\inf_{[x_{k-1}, x_k]} f = 0$ giving

$$L(f_n, P_n) = 0$$

This gives for any $\epsilon > 0$ for some $N \in \mathbb{N}$ whenever $n \geq N$ it follows

$$U(f_n, P_n) - L(f_n, P_n) = \frac{\epsilon}{2} < \epsilon$$

meaning the function is Riemann integrable with

$$\int_0^1 f_n = L(f_n) = 0$$

1.2.3 Limit

Since the sequence of values r_k are countable we have

$$\lim_{n \rightarrow \infty} f_n = \begin{cases} 1 : & x \in \mathbb{Q} \cap [0, 1] \\ 0 : & x \notin \mathbb{Q} \cap [0, 1] \end{cases} \quad (2)$$

Call this function $f(x)$

Now for any partition P since both $\mathbb{Q}$ is dense in $\mathbb{R}$ as well as $\mathbb{I}$ dense in $\mathbb{R}$ on any interval $[x_{k-1}, x_k] \in P$ there exists both a rational and irrational.

This gives $\inf_{[x_{k-1}, x_k]} f = 0$ as well $\sup_{[x_k - x_{k-1}]} f = 1$ on every interval. Which forces

$$U(f, P) = 1$$

and

$$L(f, P) = 0$$

for every partition P implying $U(f) = 1$ and $L(f) = 0$ which says that f is not Riemann integrable.

(Give measure 0 working here first and then explain how Riemann doesn't tell us of its own failure we need Lebesgue criterion, maybe start with the definition of measure 0 and briefly introduce outer measure)

1.3 Lebesgue Explanation

1.3.1 Measure 0

Definition 2 (Measure 0). A is a measure zero set if for every $\epsilon > 0$ there exists a countable covering $\bigcup_n O_n$ where each O_n is an open interval such that $A \subset \bigcup_n O_n$ where $\sum_n l(O_n) \leq \epsilon$. Where $l()$ denotes the length of a open interval defined as $b - a$ for any open interval (a, b) .

1.3.2 Lebesgue Criterion

By the Lebesgue criterion

Theorem 1 (Lebesgue Criterion). *A bounded function $f: [a, b] \rightarrow \mathbb{R}$ is Riemann integrable iff the set of discontinuities is of Measure 0*

1.3.3 f discontinuous everywhere

Recall that

$$\lim_{n \rightarrow \infty} f_n = \begin{cases} 1 & : x \in \mathbb{Q} \cap [0, 1] \\ 0 & : x \notin \mathbb{Q} \cap [0, 1] \end{cases}$$

Observe the definition of continuity.

Definition 3. A function f is continuous at a point c if for every $\epsilon > 0$ there exists a $\delta > 0$ such that whenever $|x - c| < \delta$ it follows that $|f(x) - f(c)| < \epsilon$

Now fix $\epsilon_0 \leq 1$ take $\epsilon_0 = \frac{1}{2}$. Choose any point $c \in [0, 1]$. Again both $\mathbb{Q}$ and $\mathbb{I}$ dense in $\mathbb{R}$.

First consider $c \in \mathbb{Q}$ then for any δ chosen there exists some irrational point $x \in (c - \delta, c + \delta)$ giving $|f(x) - f(c)| = 1 \geq \epsilon_0$.

Consider the case $c \in \mathbb{I}$ then for any δ chosen there exists some rational point $x \in (c - \delta, c + \delta)$ giving $|f(x) - f(c)| = 1 \geq \epsilon_0$.

Meaning the set of discontinuous points of f is all of $[0, 1]$ which has total measure 1.

To show that the set of discontinuities for every f_n is measure 0 first consider the points of discontinuity. From the section above f_n is discontinuous at

every $x \in \{r_1, r_2, \dots, r_n\}$. For any points $x \notin \{r_1, r_2, \dots, r_n\}$ we can take $\delta = \frac{\min(|x-r_1|, |x-r_2|, \dots, |x-r_n|)}{2}$ since there are a fixed amount of rationals for any fixed $n \in \mathbb{N}$. This would give us for $|x - c| < \delta \implies |f(x) - f(c)| = 0 < \epsilon$ making f_n continuous on any points $x \notin \{r_1, r_2, \dots, r_n\}$.

Now consider the open covering $\bigcup_k O_k$ such that each $O_k = (r_k - \frac{\epsilon}{2^{k+1}}, r_k + \frac{\epsilon}{2^{k+1}})$. Call D_n the set of discontinuities of f_n . Then $D_n \subset \bigcup_k O_k$ where

$$\begin{aligned} \sum_k l(O_k) &= \sum_k \frac{\epsilon}{2^k} &&= \frac{\frac{\epsilon}{2}}{1 - \frac{1}{2}} \\ &= \epsilon \end{aligned}$$

giving $\sum_k l(O_k) \leq \epsilon$ making D_n measure 0 by definition.

We now understand why each f_n integrates but f itself fails to integrate in terms of language of measure outside of the realm of the language of Riemann integration itself.

We also begin to see that the method of breaking functions up by their domain in a fixed fashion fails to work when our functions oscillate too quickly and will fail to work in more general spaces outside of $\mathbb{R}$ itself.

1.4 Lebesgue Savior